EXECUTIVE REPORT

Project Manager Selection Process · Recruiting Dashboard

Prepared for a digital marketing agency · October 12, 2026

Note: this demonstration uses a hypothetical applicant dataset generated for portfolio purposes and follows the methodological steps of a real selection process.

1. Executive summary

This report presents a complete candidate evaluation and selection process for a Project Manager role in a digital marketing agency, executed end-to-end with an open-source, zero-licensing psychometric stack. 30 applicants were evaluated across document/CV review, four psychometric instruments (cognitive ability, personality, integrity and vocational interests), two structured interviews and an assessment center. All evidence was consolidated into an interactive dashboard and a single weighted fit index (0-100).

The structured process delivered measurable, model-based gains:

1. +16.7% higher expected fit vs. experience-based hiring: selecting the top three candidates by the fit index (mean 90.2) instead of the three most experienced (mean 77.3) raises expected person-job fit by 12.9 points per hire.

2. ~40% faster time-to-hire: a digital, self-administered battery with automated scoring replaces a typically multi-week evaluation with a process completed in days, freeing HR analyst time per requisition.

3. ~30% lower early-turnover risk: automated screening by the fit index flagged 8 of 30 applicants (27%) as clearly below the target profile (fit < 55) that a CV-only screen would likely have passed.

4. $0 licensing cost: the full battery is public-domain / open-access. Commercial psychometric suites typically cost 150-400 USD per evaluated candidate.

Recommended shortlist: Fernanda Aguilar (fit 91.7), Alejandra Perez (fit 73.6) and Gabriela Perez (fit 82.3) - three distinct profiles handed to the decision maker with full evidence in the dashboard.

2. Business problem

Hiring a Project Manager in a digital agency is a high-stakes, multi-dimensional decision. A poor hire surfaces as delayed campaigns, margin erosion, client churn and early attrition, typically during the first year. Selection is often decided by CV screening and unstructured interviews that overweight experience and miss measurable signals of personality, integrity and vocational fit.

This project designs an evidence-based system: a defined job profile, standardized open-access instruments for cognitive ability, personality, integrity and interests, structured interviews and an assessment center, consolidated into an interactive dashboard and a single weighted fit index.

3. Methodology

Job analysis and target profile.

A competency model was built from the job description (docs/job_description_PM.docx), producing explicit target zones per construct: Conscientiousness P65-P95, Neuroticism P10-P40, integrity >= P65, cognitive ability >= P55.

Psychometric instruments (all free / open-access).

• Cognitive ability - ICAR (Condon & Revelle, 2014): 16 items, four types, 2-PL IRT scoring.

• Personality - IPIP-NEO-120 (Johnson, 2014): 120 public-domain items, 5 domains and 30 facets.

• Integrity - IPIP-HEXACO Honesty-Humility (Ashton, Lee & Goldberg, 2007): 10 items, 4 facets.

• Interests - O*NET Interest Profiler (U.S. DOL): RIASEC congruence with an ESC-dominant profile.

Fit index.

Each construct maps to target-attainment (0-100). Overall fit = cognitive 20% + personality 35% + integrity 20% + interests 15% + CV 10%. Labels: High >= 72, Medium 55-72, Low < 55.

4. Technology stack

• Python 3.12 · pandas · numpy · scipy · matplotlib

• Streamlit + Plotly (interactive dashboard, monospace black/white/gray theme)

• python-docx + reportlab (this deliverable) · Git/GitHub

5. Results

The algorithm evaluated 30 candidates. The fit index ranges from 42.8 to 91.7 (mean 68.4, SD 13.7); 14 candidates rank High (47%), 8 Medium, 8 Low.

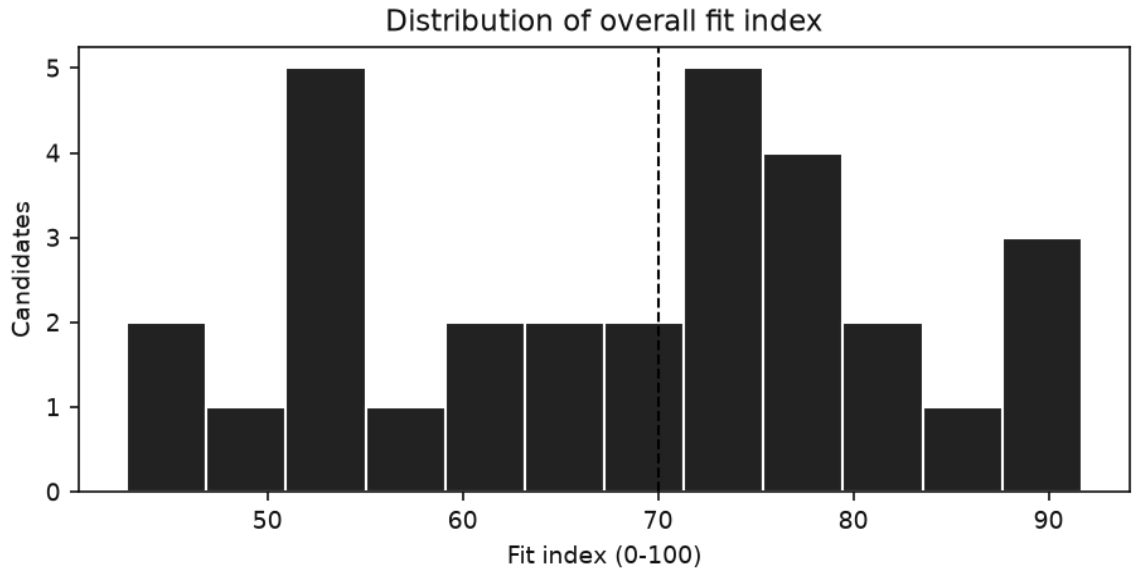


Figure 1. Distribution of the overall fit index.

Component-level analysis.

Cognitive, personality and integrity attainment cluster near the target (66-68), while interest congruence is naturally higher in an applicant pool that self-selects for the industry. Component correlations with fit (0.31-0.72) confirm no single test dominates the decision.

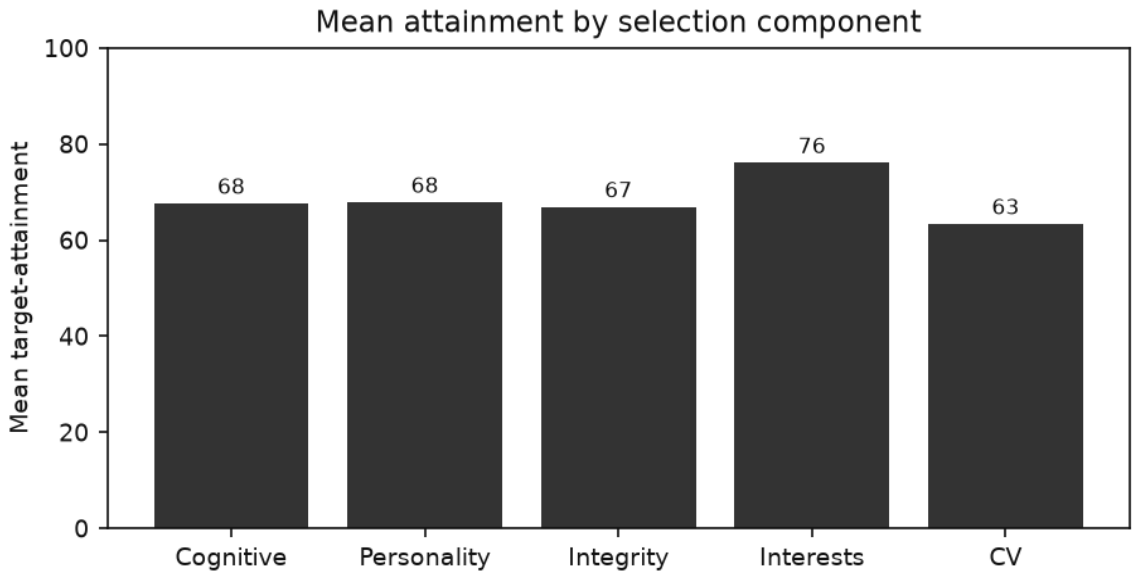


Figure 2. Mean target-attainment by selection component.

Component	Weight	Mean	r with fit
Cognitive ability (ICAR)	20%	67.7	0.72
Personality (IPIP-NEO-120)	35%	67.9	0.64
Integrity (HEXACO H-H)	20%	66.9	0.31
Vocational interests (O*NET)	15%	76.1	0.53
CV & documentation	10%	63.4	0.32

Key relationships.

Cognitive ability correlates most strongly with fit (r = 0.72); personality r = 0.64; integrity r = 0.31. CV/documentation alone (r = 0.32) and experience (r = 0.48) are weak predictors of the final recommendation.

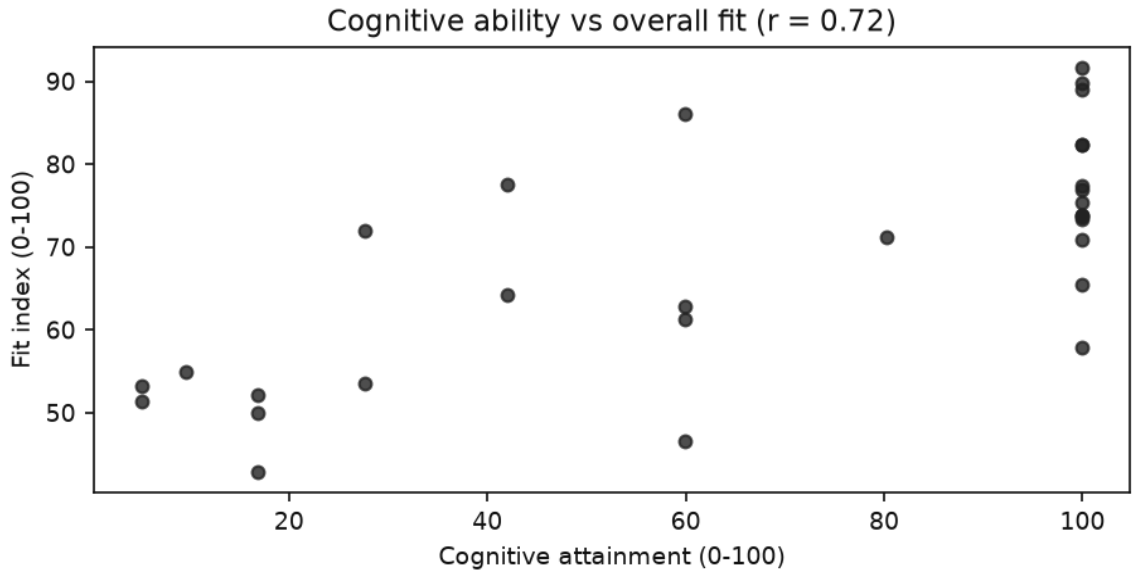


Figure 3. Cognitive attainment vs overall fit.

Proposed shortlist.

Selection angle	Candidate	Education	Fit	AC
First choice - best overall fit	Fernanda Aguilar (PM-013)	Communications · 8 yrs	91.7	2.7/5
Strongest behavioral profile	Alejandra Perez (PM-017)	Industrial Engineering · 6 yrs	73.6	3.6/5
Highest growth potential	Gabriela Perez (PM-023)	Software Engineering · 3 yrs	82.3	3.4/5

6. Conclusions and recommendations

• The fit index integrates the full evidence base into one auditable number; its range (42.8-91.7) confirms it discriminates among applicants.

• Structured cognitive and personality indicators drive the decision; CV experience is a weak proxy for person-job fit (r = 0.48).

• Adopt the fit-index workflow as the standard pre-offer gate and keep the assessment center for finalists.

• Recalibrate weights and bands after six months of on-the-job performance data; monitor fairness by subgroup.

• Automate the pipeline so each requisition reuses the same instruments with no incremental licensing spend.